NEUROLOGICALLY MOTIVATED SELF-SUPERVISED LEARNING FOR BILATERAL GAIT GEOMETRY

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INTRODUCTION

Self-supervised motion features are attractive for AI systems that must interpret people from incomplete observations. A central question is whether success at predicting a hidden feature means that the representation retains a useful physical relation. We test this question with bilateral gait geometry. Reflection exchanges left and right body landmarks and reverses a signed contrast of their movement speeds. This relation is relevant to gait asymmetry in several neurological and musculoskeletal conditions [1], while remaining independent of a diagnostic label.

MATERIALS AND METHODS

The study uses 625 quality-checked GAVD pose clips from 93 source videos [2]. Each clip is prepared independently. All 33 landmarks remain in a 64-by-33-by-3 input, and four-frame patches yield a 16-by-33 time-by-joint grid. A skeleton JEPA predicts teacher features for masked grid positions from their visible context [3]. No label, target, or affected side supervises encoder training.

We define a left-minus-right median-speed contrast over five paired joints. The null hypothesis is that JEPA does not improve its frozen ridge-regression estimate over a matched initial encoder. Five outer folds exclude complete source videos from pretraining and readout fitting; five seeds repeat each condition. Mask comparisons share starting weights, source draws, training views, and updates. We weight sources equally and resample sources, not clips, for paired intervals.

RESULTS AND DISCUSSION

Outcome	Before JEPA	After JEPA
Bilateral readout R^2	0.223	0.101–0.114
Correct-clip target preferred	33/75 checks	375/375 checks

All five trained-minus-initial contrasts were negative (-0.109 to -0.122); their saved source intervals lay below zero. Motion-weighted and connected-region masks had no clear advantage over their own count-matched random references. The predictor therefore learns within-clip feature correspondence while the tested encoder-plus-readout pipeline produces a weaker bilateral movement estimate.

The evidence has limits that matter for interpretation. Teacher feature scales differ between conditions. The expanded readout adds temporal summaries and missing-data support while increasing dimension. Moreover, preparation changes the target calculation: its agreement with the original calculation is 70.4% by sign and $R^2 = 0.218$. The study does not show that JEPA destroys geometric information, only that this training recipe and readout fail to improve this observable.

CONCLUSIONS

A latent prediction diagnostic can improve even when a prespecified geometric readout declines. Human-aware robotics may benefit from testing learned motion features against task-relevant observables before using them downstream. Future work should evaluate past-only motion prediction with persistence and velocity baselines, a stable target path, and new source or participant groups.

REFERENCES

[1] Lewek MD et al. Gait Posture 31:256–260, 2010.
[2] Ranjan R et al. IEEE Access 13:45321–45339, 2025.
[3] Abdelfattah M, Alahi A. ECCV, 2024.